Leveraging Modal Interaction and Window Dilation in Attention Network for Hyperspectral and Multispectral Remote Sensing Image Fusion

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***Abstract*—Hyperspectral and multispectral (HS-MS) image fusion aims to reconstruct high-resolution hyperspectral images (HR-HSIs) from low-resolution HSIs (LR-HSI) and HR multispectral images (HR-MSI). Convolutional neural networks (CNN) have been extensively applied to this fusion task due to their powerful feature extraction capabilities. However, the limited receptive fields of CNN-based methods make it challenging to capture long-range dependencies in images, thereby restricting their performance in fusion tasks. Attention mechanisms, which can effectively capture global correlations in data, have emerged as a popular research topic. Despite this, existing attention-based HS fusion methods have not fully integrated and coordinated the two modalities, i.e., LR-HSI and HR-MSI, resulting in limited reconstruction quality. In this article, we propose a modalities interaction and window dilation attention network (MIAN) for HS-MS image fusion, dubbed as MIAN. Specifically, our proposed MIAN introduces a modalities interaction attention block designed to explore, coordinate, and fuse the two modalities, facilitating information exchange and supplementation across different modalities. Additionally, we design a window dilation attention block that overcomes the local limitations of conventional window attention by establishing long-range connections across windows, thereby facilitating the handling of cross-scale self-similarity in remote sensing images. Furthermore, we incorporate channel attention blocks into each attention block to learn the channel dependencies which are the most important features for HS tasks. Extensive experiments demonstrate that our MIAN outperforms recent approaches on several HS datasets, achieving state-of-the-art performance.**

***Index Terms*—Hyperspectral image (HSI), image fusion, interaction attention, Transformer.**

I. Introduction

HYPERSPECTRAL images (HSI) contain hundreds of continuous narrow spectral bands spanning from the visible to the near-infrared range, enabling fine-grained characterization of the physical and chemical properties of ground materials [1]. Owing to this rich spectral information, hyperspectral (HS) tasks generally refer to a set of analytical problems that exploit both the spatial and spectral dimensions of HSIs, such as material identification, pixelwise classification, anomaly detection, target detection, and subpixel unmixing. These tasks support a wide range of real-world applications, including environmental monitoring [2], land cover classification [3], and anomaly detection in complex scenes [4]. However, due to limitations of imaging devices, an increase in spectral resolution generally leads to a decrease in spatial resolution. The low spatial resolution of HSIs can hinder their applications in various fields [5], [6]. Typically, low-resolution HSIs (LR-HSIs) and high-resolution multispectral images (HR-MSI) are acquired using different sensors. A common solution to address this issue is to fuse LR-HSIs and HR-MSIs to generate HR-HSIs [7], [8]. This process is widely known as hypersharpening [9], as it enhances the spatial resolution of HSIs by injecting the fine spatial details provided by HR-MSIs.

Over the past few decades, various HS and multispectral (HS-MS) image fusion methods have been proposed. Generally, these methods can be categorized into two types: traditional methods and deep learning (DL)-based methods. The traditional methods can further be classified into model-based methods and decomposition-based methods [10]. Model-based methods involve the manual design of priors and the optimization of objective functions to obtain HR-HSI. Such methods have been used in works like [11], [12], and [13]. On the other hand, decomposition-based methods decompose HSIs into endmembers and their corresponding abundance coefficients based on a linear spectral model, as seen in [14] and [15]. Traditional methods rely heavily on assumptions and manually designed priors, but real-world HSIs may not always conform to these assumptions.

To mitigate this limitation, DL-based methods have been proposed to tackle the HS-MS fusion problem, among which the most notable is the convolutional neural networks (CNNs)-based fusion method, including DHSIS [16], MHF-Net [17], DBIN [18], and MoG-DCN [19]. Although these CNN-based techniques have significantly improved performance compared to traditional methods, the receptive field of CNNs is limited and lacks the ability to model long-range dependencies. Consequently, CNNs may not effectively extract the comprehensive and complex features of HSIs, leading to inductive bias. To mitigate the limitations of CNN-based approaches, several works incorporate attention mechanisms or Transformers into HS-MS fusion [20], [21], [22]. While self-attention enables

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these models to capture long-range spatial dependencies, they still lack explicit mechanisms for modeling spectral dependencies and often treat all spectral bands uniformly, overlooking important interband correlations. Moreover, existing Transformer-based methods process LR-HSIs and HR-MSIs independently, making it difficult to establish cross-modal dependencies between the spectral information in LR-HSIs and the spatial information in HR-MSIs. In addition, most prior models do not jointly consider both spatial and spectral dependencies, which may result in blurred spatial structures or spectral distortions in the fused output. Furthermore, window-based attention mechanisms used to improve computational efficiency are restricted to same-scale self-similarity and thus cannot effectively capture cross-scale structural patterns commonly found in remote sensing images.

To address the aforementioned issues, we propose a novel attention network modalities interaction and window dilation attention network (MIAN) that effectively leverages the advantages of modal interaction and window dilation for HS-MS image fusion. Specifically, we introduce two key components: modalities interaction attention block (MIAB) and window dilation attention block (WDAB). MIAB takes the features of LR-HSI and HR-MSI as inputs and performs interaction attention computation. To better fuse the information from both modalities, we design two branches in MIAB: one injects LR-HSI features into HR-MSI, and the other injects HR-MSI information into LR-HSI. Moreover, WDAB applies a dilation operation to the windows, using patches of different window sizes for attention computation, overcoming the receptive field limitation of window-based attention. It enhances interaction between adjacent windows to capture more useful pixel information for computation, thus enabling the recognition of cross-scale self-similarity in remote sensing images. Additionally, we incorporate Swin Transformer blocks and channel attention blocks (CABs) into our network to strengthen feature selection and extraction. Experimental results demonstrate that our proposed MIAN outperforms other approaches, achieving state-of-the-art (SOTA) performance. In summary, our work makes three main contributions.

1) We propose MIAN for HS-MS remote sensing image fusion, which effectively extracts, aggregates, and utilizes feature information from both modalities, resulting in high-quality spatial and spectral accuracy, as well as coordinated spatial–spectral correlations in the reconstructed images.
2) We introduce MIAB, which leverages the fact that LR-HSI and HR-MSI observe the same scene to establish cross-modal correspondences. The interaction attention consists of two complementary branches: multispectral-to-hyperspectral interaction attention block (M2HIAB) injects spatial structural cues from the MSI into the HSI, while hyperspectral-to-multispectral interaction attention block (H2MIAB) propagates spectral information and spectral correlations from the HSI to the MSI.
3) We design WDAB for MIAN, which enlarges the effective receptive field of window-based attention through asymmetric dilation and enables the model to capture cross-scale spatial dependencies, allowing the network to aggregate spatial information from repeated or structurally related regions more effectively.

Experimental results on multiple remote sensing HS datasets and a natural image HS dataset demonstrate that our method achieves SOTA performance. Furthermore, extensive ablation experiments validate the effectiveness of each component in our proposed MIAN.

The article is organized as follows. Section II reviews related work on HS-MS fusion. Section III details the proposed MIAN architecture. Section IV presents experiments, including comparisons and ablation studies. Finally, Section V concludes the article.

II. Related Work

A. Traditional Methods for HS-MS Fusion

HS-MS fusion is an ill-posed problem, meaning that the obtained solution cannot be guaranteed to be consistent with the ground truth, making it highly challenging. In the past, common approaches employed priors or regularization terms to constrain the optimization equation to decompose the HSI image into more manageable elements. For example, Yokoya et al. [11] presented a coupled non-negative matrix factorization (CNMF) method for LR-HSI and HR-MSI fusion. Based on the linear spectral unmixing model, CNMF decomposes each pixel into a linear combination of pure spectral signatures (endmembers) weighted by their corresponding abundance coefficients. By coupling the factorization of both HSI and MSI, the method jointly estimates HR abundance maps from MSI and accurate endmember signatures from HSI, which are then combined to generate the HR-HSI. Simões et al. [12] bridged model-based and data-driven paradigms by combining subspace regularization with convex optimization. Lanaras et al. [13] employed physical constraints to separate spectra and performed matrix decomposition using a linear mixing model. Wei et al. [23] introduced a sparse regularization term into a series of dictionaries based on scene decomposition. Wei et al. [24] proposed a Bayesian approach that adopts gaussian mixture scale priors adopted with Gaussian mixture scale priors. Borsoi et al. [25] leveraged spectral discrepancies across images acquired at different times to address HS super-resolution problems. Dong et al. [26] presented a sparsity-based HSI super-resolution method on the basis of dictionary learning and sparse representation. Lin et al. [27] incorporated ℓ_1-norm and SSD regularizers into their model. Moreover, Chen et al. [28] employed tensor ring decomposition and factor smoothing techniques to model the high-order dependencies between HS-MS images.

In addition to these classical approaches, several recent model-driven fusion methods have further improved the efficiency and applicability of HS-MS fusion. Prévost et al. [29] proposed a Tucker-approximation-based fast fusion model that reconstructs a low-rank latent tensor with reduced computational cost. Brezini and Deville [30] employed automated extraction of endmember bundles together with sparsity-based unmixing, enabling unknown source separation without requiring training and enhancing adaptability across scenes.

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Moreover, Borsoi et al. [31] developed a coupled tensor decomposition framework capable of modeling interimage spectral variability while maintaining high computational efficiency. These recent works highlight the continued relevance of training-free, physically interpretable model-driven fusion methods.

While these traditional methods achieve favorable results in specific scenarios, they exhibit limitations in generalization ability and computational efficiency.

B. DL-Based Methods for HS-MS Fusion

Early DL-based fusion methods always employed various convolutional deformations and network connectivity strategies to construct end-to-end networks, aiming to establish the mapping relationship from LR-HSI and HR-MSI to HR-HSI. For instance, Chang et al. [32], Yuan et al. [33], and Liu et al. [34] utilized residual structures and multichannel convolutions on basic CNN architectures to learn deep features, respectively. Moreover, Han et al. [35] implemented efficient spatial and spectral information fusion within CNN architectures through joint learning strategies. Zhang et al. [36] introduced spatial edge loss and spectral edge loss to aid in the recovery of spatial and spectral information.

Integrating model-driven approaches into DL-based network represents another strategy. Such methods typically combine a feedforward process with DL priors to enhance the reconstruction performance of deep fusion networks while increasing their interpretability. For instance, Dian et al. [37] introduced a low tensor-column rank prior and utilized a deep convolutional residual network to learn image priors, constructing a minimization model based on a zero spectral distortion objective. Similarly, Dong et al. [19] unfolded traditional iterative HS super-resolution algorithms into an end-to-end denoiser, termed the MoG-DCN.

However, CNNs used in the aforementioned models exhibit limitations in modeling long-range dependencies and capturing self-similarity priors. To address these shortcomings, researchers have proposed a series of attention mechanism or Transformer-based fusion methods. For instance, Zhou et al. [38] integrated Transformers and invertible neural networks for pansharpening tasks. Hu et al. [20] applied Transformers to HSI super-resolution, leveraging spatial residual estimation to simplify training. Moreover, Hu et al. [39] proposed HSRNet in which it incorporated channel attention [40] and spatial attention modules to extract effective features from different dimensions. Deng et al. [21] proposed PSRT, utilizing a pyramid structure based on window attention to extract multiscale details. To transcend the local limitations of window-based mechanisms, they designed a Shuffle-and-Reshuffle strategy that reorganizes patches to facilitate efficient global information interaction. Jia et al. [41] proposed MSST-Net, a dual-branch network that employs spectral and spatial Transformers to separately extract global features from HSI and MSI. They incorporated multiscale embeddings and cross-modality concatenation to enhance feature richness, and further boosted performance through a self-supervised pretraining strategy involving masked autoencoders. For effective cross-modal integration, Ma et al. [42] developed the discrete cosine transform (DCT) model, which features a reciprocal fusion strategy. By employing directional pairwise multihead cross-attention, this method explicitly facilitates information transfer and interplay between the HS-MS branches. In the field of unsupervised learning, Cao et al. [22] invented an unsupervised hybrid network, combining Transformer and CNN, to solve the problem of blind HR HS-MS image fusion. In addition to the earlier studies, several recent works have made notable contributions to the field of HS-MS fusion, such as [43], [44], [45], and [46].

Due to the limitations of the receptive field, HS-MS fusion networks based on CNNs are unable to effectively extract multiscale features from HS-MS images. While attention-based methods address this issue by capturing long-range dependencies, these methods do not fully aggregate and fuse the spatial and channelwise features from the two modalities. Additionally, current visual attention mechanisms often employ window-based attention, which is effective for handling single-scale self-similarity in remote sensing images but fails to capture multiscale similarities present in the images.

III. Methodology

This section begins with an overview of the MIAN framework (see Section III-A, followed by the interaction dilation Transformer (IDT) in Section III-B and the hybrid attention group (HAG) in Section III-C. The MIAB and WDAB are then presented in Sections III-D and III-E, respectively.

A. Overall Framework

As illustrated in Fig. 1(a), the proposed MIAN is an end-to-end framework specifically designed for HS-MS image fusion, comprising three key components. Initially, shallow features are extracted and the channel dimension is enhanced via an initial convolutional layer. Subsequently, exactly T IDTs are employed to effectively extract and integrate feature information from both modalities. Finally, residual connections and convolutional layers are utilized to refine the feature representation and generate the final fused result.

Let $\boldsymbol{X} \in \mathbb{R}^{h\times w\times S}$ represent the LR-HSI, and $\boldsymbol{Y} \in \mathbb{R}^{H\times W\times s}$ represent the HR-MSI, both of which are the MIAN network's inputs. The output is the HR-HSI $\boldsymbol{Z} \in \mathbb{R}^{H\times W\times S}$. The LR-HSI has high spectral resolution with S spectral bands, but its spatial resolution is low, with h and w representing the image's height and width. In contrast, the HR-MSI has high spatial resolution, with H and W indicating the image's height and width, but its spectral resolution is only s. Additionally, $r = H/h = W/w$ is the spatial upsampling factor. At the initial stage of the network, we first use the bilinear interpolation function $\mathcal{F}_{\text{bi}}(\cdot)$ to upsample the LR-HSI, $\boldsymbol{X}$, to $\boldsymbol{M}_{\boldsymbol{X}}^{-1} \in \mathbb{R}^{H\times W\times S}$, aligning it with the HR-MSI in the spatial dimension

$$\boldsymbol{M}_{\boldsymbol{X}}^{-1} = \mathcal{F}_{\text{bi}}(\boldsymbol{X}). \tag{1}$$

Next, we use two convolutional layers as shallow feature extractors. A 3×3 convolutional layer, $\mathcal{F}_{\text{SFE1}}(\cdot)$, is employed to extract shallow features $\boldsymbol{M}_{\boldsymbol{X}}^{0} \in \mathbb{R}^{H\times W\times C}$ from $\boldsymbol{M}_{\boldsymbol{X}}^{-1}$, while another 3×3 convolutional layer, $\mathcal{F}_{\text{SFE2}}(\cdot)$, is used to extract shallow features $\boldsymbol{M}_{\boldsymbol{Y}} \in \mathbb{R}^{H\times W\times C}$ from $\boldsymbol{Y}$

$$\boldsymbol{M}_{\boldsymbol{X}}^{0} = \mathcal{F}_{\text{SFE1}}\left(\boldsymbol{M}_{\boldsymbol{X}}^{-1}\right) \tag{2}$$

$$\boldsymbol{M}_{\boldsymbol{Y}} = \mathcal{F}_{\text{SFE2}}(\boldsymbol{Y}). \tag{3}$$

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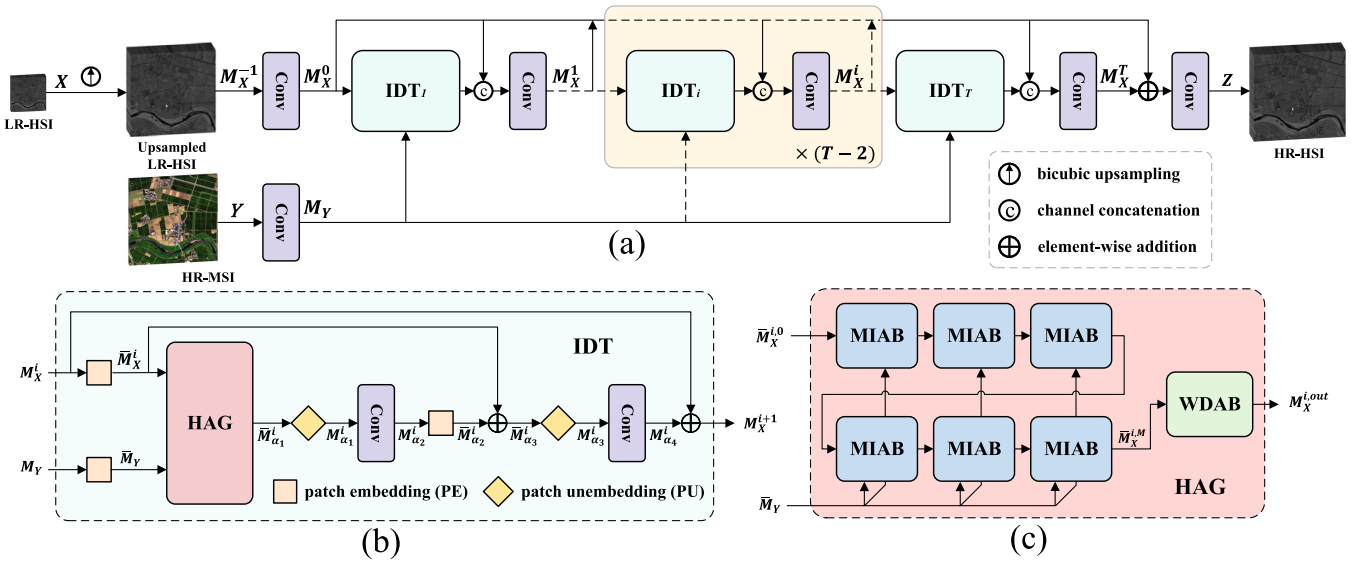


Fig. 1. Illustration of MIAN, IDT, and HAG. (a) Overall architecture of the proposed MIAN. (b) IDT. (c) HAG.

Here, C represents the number of channels in the feature map.

Subsequently, the two shallow features, $\boldsymbol{M}_X^0$ and $\boldsymbol{M}_Y$, are progressively fed into T IDTs to achieve HS-MS fusion. Since each layer at different depths of a DL network extracts distinct features and variations, we employ dense connections [47] to promote feature reuse, enable efficient information flow, enhance multidepth feature learning, and alleviate issues such as gradient vanishing and overfitting. Specifically, the input of the current IDT is the output of all previous layers. For each IDT, let the input to the ith IDT be $\boldsymbol{M}_X^{i-1}$ and the output be $\boldsymbol{M}_X^i$, with $\mathcal{F}_{\mathrm{IDT}_i}(\cdot)$ representing the function associated with the ith IDT. As a result, the dense connection of the ith IDT can be represented as follows:

$$\boldsymbol{M}_X^i = \mathrm{Conv}\left(\mathrm{Concat}\left(\boldsymbol{M}_X^0, \ldots, \boldsymbol{M}_X^{i-1}, \mathcal{F}_{\mathrm{IDT}_i}\left(\boldsymbol{M}_X^{i-1}\right)\right)\right). \tag{4}$$

Here, $i \in \{1, \ldots, T\}$, "Conv" represents a 3×3 convolutional layer, and "Concat" indicates the concatenation operation along the channel dimension. Assuming the input feature of the first IDT has a channel dimension of N_f, due to dense connections, the output channel dimension of the ith IDT increases to $i * N_f$. To reduce computational complexity and maintain model stability, after concatenation, we use a convolution layer to reduce the output channel dimension of the IDT from $i * N_f$ to N_f.

Finally, we aggregate the shallow features $\boldsymbol{M}_X^0$ and the deep features $\boldsymbol{M}_X^T$ obtained from the last IDT through a residual connection. Subsequently, a convolutional layer is applied to reconstruct the final output $\boldsymbol{Z}$

$$\boldsymbol{Z} = \mathcal{F}_{\mathrm{RE}}\left(\boldsymbol{M}_X^0 + \boldsymbol{M}_X^T\right) \tag{5}$$

where $\mathcal{F}_{\mathrm{RE}}(\cdot)$ denotes the reconstruction module, implemented using a 3×3 convolutional layer.

The proposed MIAN is trained using an ℓ_1-loss function. Given the training set $\{\boldsymbol{X}^i, \boldsymbol{Y}^i, \boldsymbol{Z}^i\}_{i=1}^N$, the loss function is defined as follows:

$$L(\Theta) = \frac{1}{N}\sum_{i=1}^{N}\left\| M_{\mathrm{MIAN}}\left(\boldsymbol{X}^i, \boldsymbol{Y}^i\right) - \boldsymbol{Z}_{\mathrm{Ground}}^i \right\|_{\ell_1} \tag{6}$$

where Θ represents all parameters of the model, and N is the total number of images in the dataset; $M_{\mathrm{MIAN}}(\boldsymbol{X}^i, \boldsymbol{Y}^i)$ and $\boldsymbol{Z}_{\mathrm{Ground}}^i$ denote the ith fused image reconstructed by MIAN and its corresponding ground truth HR-HSI, respectively.

B. Interaction Dilation Transformer

In this part, we describe the internal structure of the IDT, as shown in Fig. 1(b). Similar to the vision Transformer (ViT) [48] and the Swin Transformer [49], we incorporate a patch embedding (PE) module that transforms the 2-D image into a 1-D sequence of vectors, facilitating the further processing of features by the proposed HAG. The patch unembedding (PU) module serves as the inverse process, converting the 1-D sequence back into a 2-D feature map. The functions of the two modules are as follows:

$$\begin{aligned} \mathrm{PE}\left(\bar{\boldsymbol{M}}_{\alpha_k}^i\right) &= \boldsymbol{M}_{\alpha_k}^i, \quad k = 1, 3 \\ \mathrm{PU}\left(\boldsymbol{M}_{\alpha_k}^i\right) &= \bar{\boldsymbol{M}}_{\alpha_k}^i, \quad k = 2 \end{aligned} \tag{7}$$

where features with an overline represent 1-D vectors, while those without the overline represent 2-D matrices. After the PU module, 2-D convolution operations can be applied, with residual operations performed in both image states.

C. Hybrid Attention Group

As shown in Fig. 1(c), the proposed HAG consists of M MIAB and one WDAB. The MIAB serves to fuse the features of HS-MS images. By iteratively applying the MIAB, we can effectively explore the correlations between the HS-MS images, ensuring that the resulting HR-HSI conforms to the distribution of the reference real image set in terms of spatial,

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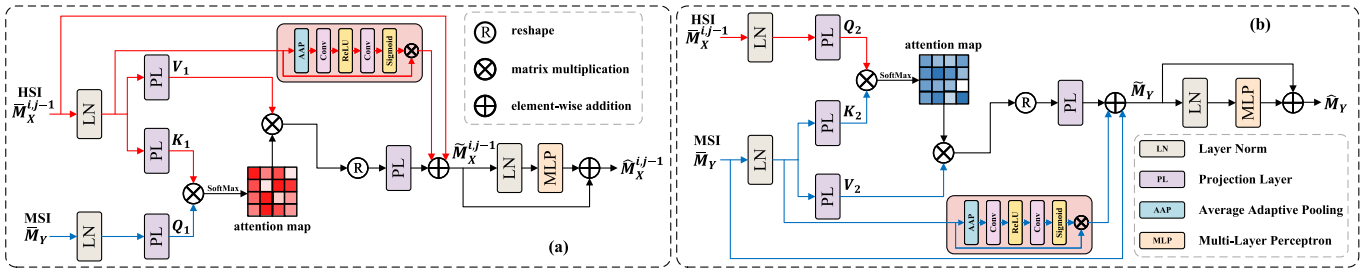


Fig. 2. Illustration of (a) M2HIAB and (b) H2MIAB.

channel, and overall feature correlations. The operation for the jth MIAB module can be expressed as follows:

$$\bar{\boldsymbol{M}}_X^{i,j} = \mathcal{F}_{\mathrm{MIAB}}\left(\bar{\boldsymbol{M}}_X^{i,j-1}, \bar{\boldsymbol{M}}_Y\right) \tag{8}$$

where $\mathcal{F}_{\mathrm{MIAB}}(\cdot)$ is the corresponding function of the MIAB module, $\bar{\boldsymbol{M}}_X^{i,j-1}$ and $\bar{\boldsymbol{M}}_X^{i,j}$ represent the input and output of the jth MIAB module for the MSI features, and $\bar{\boldsymbol{M}}_Y$ is the HS feature after passing through the patch embed module.

The WDAB module overcomes the limitations of the Swin Transformer's window attention mechanism by utilizing a larger pixel range, thereby enabling cross-window information interaction, which can be represented as follows:

$$\boldsymbol{M}_X^{i,\mathrm{out}} = \mathcal{F}_{\mathrm{WDAB}}\left(\bar{\boldsymbol{M}}_X^{i,M}\right) \tag{9}$$

where $\bar{\boldsymbol{M}}_X^{i,M}$ denotes the output of the Mth module, i.e., the last MIAB module, $\boldsymbol{M}_X^{i,\mathrm{out}}$ represents the output of the HAG, and $\mathcal{F}_{\mathrm{WDAB}}(\cdot)$ is the function corresponding to the WDAB module.

D. Modalities Interaction Attention Block (IAB)

The proposed MIAB is composed of two IABs and one Swin Transformer block. Given that HS-MS images have more spectral channels than typical RGB images, we integrate a CAB into each attention module to identify which spectral channels contain more important information. This design allows the enhancement of key channel information by allocating higher weights to them while suppressing irrelevant channels. As shown in Fig. 2, two IABs in MIAB are responsible for different directions of information interaction: the M2HIAB injects spatial information, including structure and texture, from the MSI into the HSI; whereas the hyperspectral-to-multispectral IAB (H2MIAB) transfers spectral information and spectral correlations from the HSI to the MSI. Note that feature flows from the HSI modality are marked with red lines, while those from the MSI modality are marked with blue lines. The outputs from both blocks are fused via a residual connection. The fused features are then passed into the channel attention Swin Transformer block (CSTB), which contains a window-based multihead self-attention module to capture local feature relationships within the window, while the CAB is responsible for allocating feature correlations along the channel dimension. Through this combination, MIAB simultaneously models three types of dependencies essential for HS-MS fusion: spatial dependencies conveyed through the M2HIAB branch, spectral dependencies emphasized by the CAB modules, and cross-modal dependencies established by the bidirectional interaction between LR-HSI and HR-MSI. This unified design enables MIAB to effectively integrate complementary information from both modalities while preserving structural details and spectral fidelity

$$\begin{aligned}\boldsymbol{M}_X^{i,j} &= \mathcal{F}_{\mathrm{MIAB}}\left(\bar{\boldsymbol{M}}_X^{i,j-1}, \bar{\boldsymbol{M}}_Y\right) \\ &= \mathcal{F}_{\mathrm{CSTB}}\left(\mathcal{F}_{\mathrm{M2HIAB}}\left(\bar{\boldsymbol{M}}_X^{i,j-1}, \bar{\boldsymbol{M}}_Y\right)\right. \\ &\quad \left. + \mathcal{F}_{\mathrm{H2MIAB}}\left(\bar{\boldsymbol{M}}_X^{i,j-1}, \bar{\boldsymbol{M}}_Y\right)\right)\end{aligned} \tag{10}$$

where $\mathcal{F}_{\mathrm{MIAB}}(\cdot)$ represents MIAB. $\mathcal{F}_{\mathrm{M2HIAB}}(\cdot)$, $\mathcal{F}_{\mathrm{H2MIAB}}(\cdot)$, and $\mathcal{F}_{\mathrm{CSTB}}(\cdot)$ represent the M2HIAB, H2MIAB, and CSTB modules, respectively.

1) Multispectral-to-Hyperspectral IAB: This branch facilitates the mutual transfer of geometric features, structural details, and texture information between the HSI and MSI in the spatial dimensions. Given the flattened inputs $\bar{\boldsymbol{M}}_X^{i,j-1} \in \mathbb{R}^{L\times C}$ and $\bar{\boldsymbol{M}}_Y \in \mathbb{R}^{L\times C}$ (where $L = H \times W$), M2HIAB first partitions these into nonoverlapping window patches $\{\boldsymbol{P}_{X_1}^1, \boldsymbol{P}_{X_1}^2, \ldots, \boldsymbol{P}_{X_1}^U\}$ and $\{\boldsymbol{P}_{Y_1}^1, \boldsymbol{P}_{Y_1}^2, \ldots, \boldsymbol{P}_{Y_1}^U\}$. Here, $\boldsymbol{P}_{X_1}^u \in \mathbb{R}^{l\times C}(u = 1, 2, \ldots, U)$ and $\boldsymbol{P}_{Y_1}^u \in \mathbb{R}^{l\times C}(u = 1, 2, \ldots, U)$, where $U = L/l$ is the total number of patches, and l denotes the length of each patch. Next, $\boldsymbol{P}_{X_1}^u(u = 1, 2, \ldots, U)$ is linearly mapped to keys $\boldsymbol{K_1} \in \mathbb{R}^{l\times d}$ and values $\boldsymbol{V_1} \in \mathbb{R}^{l\times d}$, while $\boldsymbol{P}_{Y_1}^u(u = 1, 2, \ldots, U)$ is linearly mapped to queries $\boldsymbol{Q_1} \in \mathbb{R}^{l\times d}$, as shown below

$$\begin{cases}\boldsymbol{Q_1} = \boldsymbol{P}_{Y_1}^u \boldsymbol{W}^{Q_1} \\ \boldsymbol{K_1} = \boldsymbol{P}_{X_1}^u \boldsymbol{W}^{K_1} \\ \boldsymbol{V_1} = \boldsymbol{P}_{X_1}^u \boldsymbol{W}^{V_1}\end{cases} \tag{11}$$

where $\boldsymbol{W}^{Q_1}$, $\boldsymbol{W}^{K_1}$, and $\boldsymbol{W}^{V_1} \in \mathbb{R}^{C\times d}$ are the corresponding mapping matrices.

Formally, the multispectral-to-hyperspectral interaction attention (M2HIA) is computed as follows:

$$\begin{aligned}&\mathrm{M2HIA}\left(\boldsymbol{Q_1}, \boldsymbol{K_1}, \boldsymbol{V_1}\right) \\ &= \mathrm{SoftMax}\left(\boldsymbol{Q_1}\boldsymbol{K_1}^T / \sqrt{d} + \boldsymbol{B}\right)\boldsymbol{V_1}\end{aligned} \tag{12}$$

where $\boldsymbol{K_1}^T$ represents the transpose of $\boldsymbol{K_1}$, and $\boldsymbol{B}$ denotes the learnable relative positional encoding. Following the approach in [49], we alternate between conventional and shifted window partitioning constructions. The input $\bar{\boldsymbol{M}}_X^{i,j-1}$ is aggregated with the feature map from the CAB and the results generated by the interaction attention through residual connections to

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obtain the intermediate output $\tilde{\boldsymbol{M}}_X^{i,j-1}$. Additionally, to transform the features, as in other Transformers, a multilayer perceptron (MLP) is applied after the attention calculation to obtain the final output of this module, $\hat{\boldsymbol{M}}_X^{i,j-1}$. The MLP consists of two fully connected layers, with the nonlinear activation function Gaussian error linear unit (GELU) applied between these layers. Moreover, the LayerNorm is applied before both M2HIA and the MLP to balance the feature distributions. Furthermore, residual connections are also used after the MLP. The entire process of M2HIAB can be expressed as follows:

$$\hat{\boldsymbol{M}}_X^{i,j-1} = \mathcal{F}_{\text{M2HIAB}}\left(\bar{\boldsymbol{M}}_X^{i,j-1}, \bar{\boldsymbol{M}}_Y\right) \tag{13}$$

where

$$\begin{aligned}\tilde{\boldsymbol{M}}_X^{i,j-1} &= \text{M2HIA}\left(\text{LN}\left(\bar{\boldsymbol{M}}_Y\right), \text{LN}\left(\bar{\boldsymbol{M}}_X^{i,j-1}\right)\right) \\ &\quad + \bar{\boldsymbol{M}}_X^{i,j-1} + \text{CAB}\left(\text{LN}\left(\bar{\boldsymbol{M}}_X^{i,j-1}\right)\right) \\ \hat{\boldsymbol{M}}_X^{i,j-1} &= \text{MLP}\left(\text{LN}\left(\tilde{\boldsymbol{M}}_X^{i,j-1}\right)\right) + \tilde{\boldsymbol{M}}_X^{i,j-1}. \end{aligned} \tag{14}$$

2) Hyperspectral-to-Multispectral IAB: This branch is designed to coordinate, synchronize, and fuse the spectral information from HSI and MSI. Given the flattened inputs $\bar{\boldsymbol{M}}_X^{i,j-1} \in \mathbb{R}^{L\times C}$ and $\bar{\boldsymbol{M}}_Y \in \mathbb{R}^{L\times C}$, similar to M2HIAB, H2MIAB first divides them into nonoverlapping window patches $\{\boldsymbol{P}_{X_2}^1, \boldsymbol{P}_{X_2}^2, \ldots, \boldsymbol{P}_{X_2}^U\}$ and $\{\boldsymbol{P}_{Y_2}^1, \boldsymbol{P}_{Y_2}^2, \ldots, \boldsymbol{P}_{Y_2}^U\}$. Then, $\boldsymbol{P}_{Y_2}^u (u = 1, 2, \ldots, U)$ is linearly mapped into key $\boldsymbol{K_2} \in \mathbb{R}^{l\times d}$ and value $\boldsymbol{V_2} \in \mathbb{R}^{l\times d}$, while $\boldsymbol{P}_{X_2}^u (u = 1, 2, \ldots, U)$ is linearly mapped into query $\boldsymbol{Q_2} \in \mathbb{R}^{l\times d}$. The entire process of H2MIAB can be expressed as follows:

$$\hat{\boldsymbol{M}}_Y = \mathcal{F}_{\text{H2MIAB}}\left(\bar{\boldsymbol{M}}_X^{i,j-1}, \bar{\boldsymbol{M}}_Y\right) \tag{15}$$

where

$$\begin{aligned}\tilde{\boldsymbol{M}}_Y &= \text{H2MIA}\left(\text{LN}\left(\bar{\boldsymbol{M}}_X^{i,j-1}\right), \text{LN}\left(\bar{\boldsymbol{M}}_Y\right)\right) \\ &\quad + \bar{\boldsymbol{M}}_Y + \text{CAB}\left(\text{LN}\left(\bar{\boldsymbol{M}}_Y\right)\right) \\ \hat{\boldsymbol{M}}_Y &= \text{MLP}\left(\text{LN}\left(\tilde{\boldsymbol{M}}_Y\right)\right) + \tilde{\boldsymbol{M}}_Y. \end{aligned} \tag{16}$$

Here, $\tilde{\boldsymbol{M}}_Y$ is the intermediate output, and $\hat{\boldsymbol{M}}_Y$ is the final output. H2MIA represents the hyperspectral-to-multispectral interaction attention operation.

Now that we have obtained the outputs $\hat{\boldsymbol{M}}_X^{i,j-1}$ from the M2HIAB branch and $\hat{\boldsymbol{M}}_Y$ from the H2MIAB branch, we first add the results from both branches to fuse the information from the two branches. Subsequently, we use the CSTB to further fuse and enhance the features, which can be expressed as follows:

$$\boldsymbol{M}_X^{i,j} = \mathcal{F}_{\text{CSTB}}\left(\hat{\boldsymbol{M}}_X^{i,j-1} + \hat{\boldsymbol{M}}_Y\right) \tag{17}$$

where

$$\begin{aligned}\ddot{\boldsymbol{M}}_X^{i,j-1} &= \text{MSA}\left(\text{LN}\left(\hat{\boldsymbol{M}}_X^{i,j-1}\right)\right) \\ &\quad + \hat{\boldsymbol{M}}_X^{i,j-1} + \text{CAB}\left(\text{LN}\left(\hat{\boldsymbol{M}}_X^{i,j-1}\right)\right) \\ \boldsymbol{M}_X^{i,j} &= \text{MLP}\left(\text{LN}\left(\ddot{\boldsymbol{M}}_X^{i,j-1}\right)\right) + \ddot{\boldsymbol{M}}_X^{i,j-1} \end{aligned} \tag{18}$$

where $\ddot{\boldsymbol{M}}_X^{i,j-1}$ represents the intermediate feature representation in the CSTB module after processing with the window attention mechanism, while $\boldsymbol{M}_X^{i,j}$ denotes the final output of the CSTB module, which also serves as the output of the entire MIAB module.

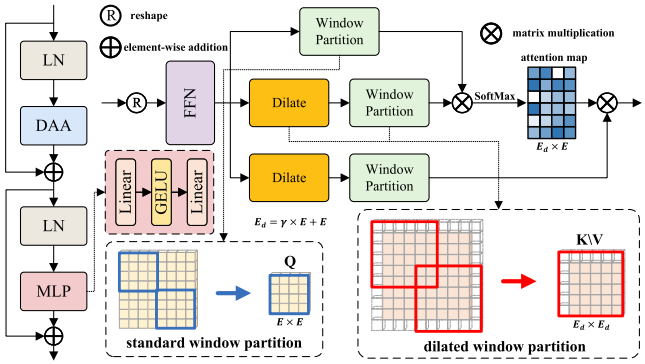


Fig. 3. Illustration of WDAB.

E. Window Dilation Attention Block

We introduce a WDAB to directly establish cross-window connections and enhance the expressive power of window attention. Our proposed WDAB consists of a dilated asymmetric attention (DAA) layer and an MLP layer. The overall architecture of WDAB is similar to a standard Swin Transformer block. However, for the DAA, as depicted in Fig. 3, we use different window sizes to partition the input features.

The result $\bar{\boldsymbol{M}}_X^{i,M}$ after passing through M MIABs serves as the input to the WDAB. Upon entering the WDAB, $\bar{\boldsymbol{M}}_X^{i,M} \in \mathbb{R}^{L\times C}$ is reshaped into $\boldsymbol{M}_X^{i,M} \in \mathbb{R}^{H\times W\times C}$. Then, $\boldsymbol{M}_X^{i,M}$ is linearly mapped to $\boldsymbol{Q_3}, \boldsymbol{K_3}, \boldsymbol{V_3} \in \mathbb{R}^{H\times W\times C}$. $\boldsymbol{Q_3}$ is partitioned into $(HW)/(E^2)$ nonoverlapping window patches of size $E\times E$. Meanwhile, dilation is applied to $\boldsymbol{K_3}$ and $\boldsymbol{V_3}$, by padding zero pixels around the feature with a width of $(\gamma E)/2$. Then, $\boldsymbol{K_3}$ and $\boldsymbol{V_3}$ are partitioned into $(HW)/(E^2)$ overlapping window patches of size $E_d \times E_d$. The two types of window patches are asymmetric, and their relationship can be expressed as follows:

$$E_d = \gamma \times E + E \tag{19}$$

where γ is the hyperparameter controlling the dilation rate. In our experiments, we set $\gamma = 0.5$. From a programming perspective, the operation of DAA can be interpreted as follows: standard window partitioning is equivalent to a sliding window operation where both the kernel size and stride are equal to E. In contrast, dilated window partitioning can be regarded as a sliding window operation with a stride of E but a kernel size of E_d. The computation of the DAA is given by

$$\begin{aligned}&\text{DAA}\left(\boldsymbol{Q_3}, \boldsymbol{K_3}, \boldsymbol{V_3}\right) \\ &= \text{SoftMax}\left(\boldsymbol{Q_3}\boldsymbol{K_3}^T / \sqrt{d} + \boldsymbol{B_d}\right)\boldsymbol{V_3} \end{aligned} \tag{20}$$

where $\boldsymbol{B_d} \in \mathbb{R}^{E_d\times E}$ represents the relative position encoding adjusted for DAA.

By enlarging the effective receptive field through asymmetric dilated windows, WDAB establishes cross-window interactions and captures cross-scale spatial dependencies that recur in remote sensing images. This design allows the network to aggregate spatial information from repeated or

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structurally related regions more effectively, thereby enhancing the expressive power of window-based attention.

IV. Experimental Results and Analyses

A. Experimental Settings

1) Datasets: We conduct experiments under two spatial scales, ×4 and ×32. For the ×4 scale, we select two HS remote sensing datasets, including the Chikusei dataset [50] and the Pavia Centre dataset [51], as well as a natural HSI dataset, Columbia Aerial Videography and Photography Experiment (CAVE) [52]. For the ×32 scale, we utilize three HS remote sensing datasets: the Pavia University dataset [51], the Chikusei dataset [50], and the Xiongan dataset [53]. All degradation operations follow Wald's protocol [54], including Gaussian blur-based spatial downsampling for generating LR-HSI and SRF-based spectral projection for synthesizing HR-MSI.

The Chikusei dataset contains HSIs (128 bands, 363–1018 nm) of urban/rural scenes in Ibaraki, Japan, with a resolution of 2.5 m/pixel (2517 × 2335 pixels). Following standard practice, we remove several water vapor absorption bands and noisy bands before use. The Pavia datasets (ROSIS sensor, Italy) include Pavia Centre (102 bands, 1096 × 1096 pixels, 1.3 m resolution) and Pavia University (93 bands, 610 × 340 pixels, 430–860 nm). Similar to prior HS-MS fusion works, heavily corrupted or extremely noisy bands are discarded during preprocessing. The Xiongan dataset (Gaofen project, China) covers 400–1000 nm (downsampled to 93 bands) at 0.5 m resolution (3750 × 1580 pixels). For generalization, the CAVE dataset (natural images) provides 31 bands (400–700 nm, 512 × 512 pixels).

For ×4 scale, the training data is extracted as follows: Chikusei uses a 1000 × 2000 top-left region cropped into 64×64 HR-HSI patches ground truth (GT), with corresponding 64 × 64 × 3 HR-MSI and 16 × 16 × 128 LR-HSI patches, while six 680 × 680 × 128 patches serve as test data; Pavia Centre employs the upper half for 64 × 64 training patches and four 256 × 256 test patches; and CAVE utilizes 20 images (3920 patches of 64 × 64 × 31 HR-HSI) processed with Gaussian blur (σ = 0.5) and ×4 downsampling for LR-HSI, plus Nikon D700 spectral response for HR-MSI, split 80%/20% for training/validation, with remaining 12 images for testing—similar procedures apply to other datasets. For all ×4 experiments, HR-MSI is generated by applying the corresponding MS sensor spectral response functions (SRFs) to the ground-truth HR-HSI: the Nikon D700 SRF is used for CAVE, while the airborne sensor SRFs are used for Chikusei and Pavia datasets.

At the ×32 scale, we follow the data processing method from [22]. For the Pavia University dataset, the region in the lower left corner with a resolution of 320 × 320 is used as the test set. For the Chikusei and Xiongan datasets, we select a 512 × 512 region from the top-left corner as the test image. The remaining portions of the three datasets are cropped into overlapping 128×128 patches to form the training set. According to [55], the images in the dataset are used as GT. For all ×32 experiments, HR-MSI is synthesized by applying an IKONOS-like spectral response filter to the HR-HSI following the standard Wald protocol. Spatial degradation is simulated to generate LR-HSI by applying Gaussian blur-based downsampling, with a Gaussian kernel size of 34 and a standard deviation of 28.

TABLE I
Implementation Details and Hyperparameters

Parameter	Value
Implementation Environment	
Framework	PyTorch
Hardware	Single NVIDIA RTX A6000 GPU
Training Settings	
Optimizer	AdamW
Learning Rate	10^{-4}
Weight Decay	10^{-4}
Epochs	2000
Batch Size	16
Model Hyperparameters	
Number of IDTs (T)	3
Number of MIABs (M)	6
Window Size	8
Dilation Coefficient (γ)	0.5

2) Methods of Comparison: We select several DL-based HSI fusion methods and a DL-based pansharpening method for comparison with our approach. These methods include CNN-based methods: MSDCNN [33], SSF-CNN [35], PDCon-SSF [35], TFNet [34], ResTFNet [34], and SSR-NET [36]; Transformer/attention-based methods: PSRT [21], MSST-Net [41], and DCTransformer (DCT) [42]; and a deep unfolding method: MoG-DCN [19]. We train these comparison algorithms using the open-source code provided by the authors in the respective papers. It is important to note that the training and testing setups are kept consistent across all methods.

3) Evaluation Metrics: In this section, we select six metrics to quantitatively assess the performance of all methods. These include peak signal-to-noise ratio (PSNR) [56], spectral angle mapper (SAM) [57], relative global dimensionless error in synthesis (ERGAS) [58], structural similarity index (SSIM) [56], root mean square error (RMSE) [59], and correlation coefficient (CC) [59]. Among these metrics, higher values of PSNR, SSIM, and CC indicate better model performance, while lower values of SAM, ERGAS, and RMSE suggest better performance. In the experimental results, the best and second-best values are highlighted in red and blue, respectively. In the case of ties for the best performance, all matching values are marked in red.

4) Implementation Details: The implementation details and hyperparameters are summarized Table I. We train MIAN using the AdamW optimizer, with a learning rate set to 10^{-4} and a weight decay coefficient set to 10^{-4}. The training process is configured for 2000 epochs with a batch size of 16. We use PyTorch to train and test our model. All experiments are conducted on a single NVIDIA RTX A6000 GPU. Based on the results of the ablation experiments, we set the number of IDTs T to 3, the number of MIABs M to 6, the window size in the attention mechanism to 8, and the dilation coefficient γ

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TABLE II
COMPARISON OF OUR METHOD WITH OTHER METHODS ON SIX EVALUATION METRICS AT ×4 SCALE

Dataset	Metric	MSDCNN [33]	SSF-CNN [35]	PDCon-SSF [35]	TFNet [34]	ResTFNet [34]	SSR-NET [36]	MoG-DCN [19]	MSST-Net [41]	PSRT [21]	DCT [42]	MIAN (Our Method)
Chikusei	PSNR ↑	31.3034	31.5082	26.8743	29.7276	29.7226	31.1687	29.5258	30.3869	30.7266	**31.5188**	**31.7995**
	SAM ↓	2.0458	1.9982	6.4127	2.2816	2.2244	2.1475	2.2525	2.2214	2.3861	**1.9338**	**1.8900**
	ERGAS ↓	3.0653	3.0005	6.5097	3.3289	3.3273	3.1746	3.3280	3.1849	3.5654	**2.9161**	**2.8673**
	SSIM ↑	0.9642	0.9654	0.9248	0.9511	0.9514	0.9628	0.9477	0.9543	0.9579	**0.9648**	**0.9669**
	RMSE ↓	0.0096	**0.0093**	0.0158	0.0115	0.0115	0.0097	0.0117	0.0106	0.0102	0.0094	**0.0091**
	CC ↑	0.9875	**0.9883**	0.9677	0.9821	0.9818	0.9874	0.9818	0.9849	0.9859	0.9882	**0.9890**
Pavia Centre	PSNR ↑	45.1628	45.2034	32.8066	45.8495	45.9646	44.7637	45.8262	46.8903	46.1025	**47.0102**	**47.7005**
	SAM ↓	2.1238	2.1083	5.8142	2.0590	2.0385	2.2046	2.0166	1.9279	2.0166	**1.9010**	**1.8197**
	ERGAS ↓	1.1455	1.1388	3.5328	1.1135	1.0941	1.1808	1.0865	1.0409	1.0697	**1.0100**	**0.9563**
	SSIM ↑	0.9964	0.9964	0.9852	0.9967	0.9968	0.9962	0.9967	0.9971	0.9968	**0.9972**	**0.9974**
	RMSE ↓	0.0036	0.0036	0.0159	0.0033	0.0033	0.0038	0.0033	0.0029	0.0032	**0.0029**	**0.0026**
	CC ↑	0.9989	0.9989	0.9846	0.9990	0.9990	0.9988	0.9990	0.9991	0.9991	**0.9992**	**0.9992**
CAVE	PSNR ↑	42.7545	39.5026	39.9218	44.8064	45.0455	40.0702	46.5731	47.1089	45.1252	**47.8420**	**48.5334**
	SAM ↓	2.4119	3.2289	3.7326	1.9453	1.9276	3.3271	1.7799	1.6944	1.9870	**1.5435**	**1.4433**
	ERGAS ↓	1.7023	2.3401	2.4746	1.4257	1.4032	2.2060	1.2016	1.1629	1.3438	**1.0609**	**0.9983**
	SSIM ↑	0.9965	0.9937	0.9932	0.9976	0.9977	0.9943	0.9982	0.9984	0.9976	**0.9986**	**0.9987**
	RMSE ↓	0.0032	0.0048	0.0046	0.0024	0.0023	0.0045	0.0019	0.0018	0.0023	**0.0016**	**0.0015**
	CC ↑	0.9966	0.9938	0.9931	0.9972	0.9972	0.9941	0.9975	0.9977	0.9972	**0.9979**	**0.9980**

TABLE III
COMPARISON OF OUR METHOD WITH OTHER METHODS ON SIX EVALUATION METRICS AT ×32 SCALE

Dataset	Metric	MSDCNN [33]	SSF-CNN [35]	PDCon-SSF [35]	TFNet [34]	ResTFNet [34]	SSR-NET [36]	MoG-DCN [19]	MSST-Net [41]	PSRT [21]	DCT [42]	MIAN (Our Method)
Pavia University	PSNR ↑	52.3730	53.1393	53.0192	50.3053	50.1830	52.3017	51.3705	51.4515	50.5674	**53.8647**	**54.1242**
	SAM ↓	1.5249	1.4547	1.4547	1.8696	1.8745	1.5686	1.7521	2.0847	1.6203	**1.3462**	**1.3244**
	ERGAS ↓	0.4939	0.4640	0.4692	0.6068	0.6147	0.5130	0.5251	0.5486	0.5443	**0.4453**	**0.4401**
	SSIM ↑	0.9977	0.9980	0.9980	0.9962	0.9960	0.9975	0.9971	0.9970	0.9969	**0.9983**	**0.9983**
	RMSE ↓	0.0029	0.0027	0.0027	0.0036	0.0037	0.0029	0.0032	0.0035	0.0034	**0.0025**	**0.0025**
	CC ↑	0.9936	0.9942	0.9942	0.9905	0.9907	0.9932	0.9918	0.9871	0.9927	**0.9951**	**0.9953**
Chikusei	PSNR ↑	44.9468	44.7980	44.8394	42.2427	42.7466	45.1949	42.3777	39.3245	44.6889	**46.2044**	**46.2852**
	SAM ↓	1.4789	1.4856	1.4673	1.8685	1.8046	1.4541	1.8968	2.6654	1.5506	**1.3159**	**1.2980**
	ERGAS ↓	0.2618	0.2691	0.2668	0.3566	0.3339	0.2546	0.3393	0.4523	0.2645	**0.2216**	**0.2210**
	SSIM ↑	0.9974	0.9973	0.9973	0.9952	0.9957	0.9975	0.9956	0.9921	0.9972	**0.9980**	**0.9981**
	RMSE ↓	0.0060	0.0061	0.0060	0.0080	0.0076	0.0058	0.0079	0.0112	0.0062	**0.0053**	**0.0052**
	CC ↑	0.9869	0.9869	0.9872	0.9798	0.9814	0.9878	0.9822	0.9595	0.9860	**0.9905**	**0.9904**
Xiongan	PSNR ↑	52.7399	52.7338	52.7616	52.1825	52.1348	52.3180	52.1091	51.2211	53.1958	**53.2978**	**53.4306**
	SAM ↓	0.7084	0.7080	0.7025	0.7785	0.7840	0.7533	0.7464	0.8721	0.6559	**0.6466**	**0.6356**
	ERGAS ↓	0.0899	0.0895	0.0891	0.0980	0.0983	0.0969	0.0971	0.1092	0.0830	**0.0818**	**0.0802**
	SSIM ↑	0.9978	0.9978	0.9979	0.9975	0.9975	0.9975	0.9974	0.9971	0.9981	**0.9982**	**0.9982**
	RMSE ↓	0.0028	0.0028	0.0028	0.0031	0.0031	0.0030	0.0030	0.0034	0.0026	**0.0025**	**0.0025**
	CC ↑	0.9990	0.9990	0.9990	0.9988	0.9988	0.9988	0.9989	0.9985	0.9991	**0.9992**	**0.9992**

to 0.5. To ensure a consistent evaluation, we use the same set of training hyperparameters for all datasets and scales, with the batch size, total epochs, and learning rate fixed. The only adjustment involves the input layer dimensions, which are set to match the specific number of spectral bands of each dataset.

B. Experimental Results

1) Results of Comparison Experiments: In this section, we compare and analyze the results of our method with other methods mentioned above.

a) Comparison of evaluation metrics: We compare our proposed MIAN with different DL-based fusion methods. The results for the Chikusei, Pavia Centre, and CAVE datasets at the ×4 scale are presented in Table II. The metrics for the Pavia University, Chikusei, and Xiongan datasets at the ×32 scale are shown in Table III. In the table, the best result for each metric is highlighted in red bold, and the second-best result is highlighted in blue bold. From the table, it can be seen that our results achieve the best metric results across both scales and all datasets, with only one data point slightly lower than DCT.

Table II presents the results of all methods at the ×4 scale. Specifically, on the Chikusei dataset, the average PSNR of our MIAN is 0.3 dB higher than that of the second-best network, DCT, and the average RMSE is 3.2% lower than that of the second-best SSF-CNN. On the Pavia Centre dataset, the average PSNR of MIAN is 0.7 dB higher than that of the second-best network, DCT, and the average RMSE is 10.3% lower than that of DCT. On the CAVE dataset, the proposed

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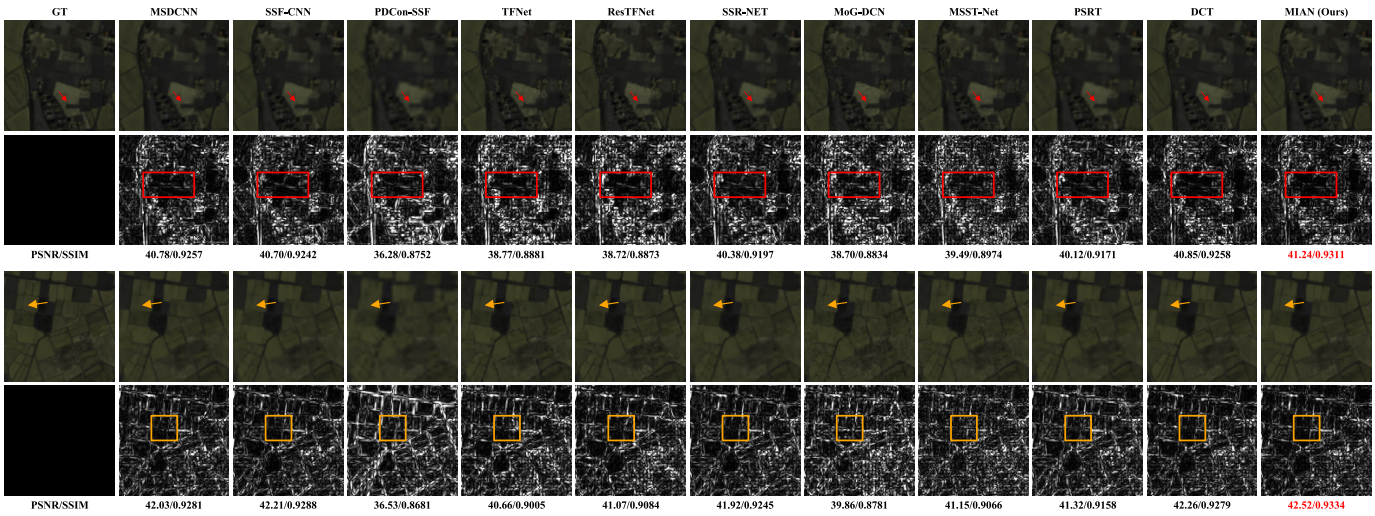


Fig. 4. Reconstructed pseudo-color images and error maps of various methods on selected regions of the Chikusei test set at $\times 4$ scale.

MIAN achieves a PSNR that is 0.7 dB higher than that of the second-best network, DCT, and the average SAM is 6.5% lower than that of DCT.

As shown in Table III, despite the challenge of fusion at the extremely large magnification factor of $\times 32$, our method still performs excellently. Specifically, on the Pavia University dataset, the average PSNR of our MIAN is 0.3 dB higher than that of the second-best network, DCT, and the average RMSE is 2.1% lower than that of DCT. On the Chikusei dataset, the average PSNR of MIAN is 0.1 dB higher than that of the second-best network, DCT, and the average RMSE is 1.5% lower than that of DCT. On the Xiongan dataset, the proposed MIAN achieves a PSNR that is 0.7 dB higher than that of the second-best network, DCT, and the average SAM is 2.3% lower than that of DCT. While the improvement over the strong Transformer-based DCT is sometimes narrow, MIAN demonstrates superior overall performance across diverse datasets. This confirms that our proposed window dilation and interaction mechanisms provide a robust advantage in capturing complex cross-scale spatial–spectral dependencies.

In addition to evaluating the quality of the fused images from a quantitative perspective using numerical metrics, we also present visual images that reveal artifacts and distortions that cannot be uncovered by quantitative analysis alone. We display pseudo-color images (which convert MSIs into visual approximations of RGB images), error maps, heatmaps, and spectral response curves to validate the superiority of our proposed MIAN.

b) Pseudo-color images and error maps: We extract three spectral bands corresponding to the red, green, and blue wavelengths from the HSI to construct pseudo-color images. The difference between the pseudo-color images of the GT and the reconstructed images is then taken, followed by absolute value transformation to generate error maps in grayscale. At the $\times 4$ scale, we compare the restoration images generated by the models listed in Section IV-A2 with our MIAN and display the GTs as References.

For the Chikusei dataset, we select regions 2 and 6 of the test image for display. We use bands 72, 81, and 88 to form pseudo-color images. The fusion results are shown in Fig. 4, where the first and third rows display the pseudo-color images of regions 2 and 6, respectively, and the second and fourth rows present the error maps for better subjective visual evaluation. We also calculate the PSNR and SSIM values for each image compared to the GT. From the pseudo-color images, it can be observed that compared to other methods, our method produces better fusion results. Specifically, in region 2, at the location indicated by the red arrow on the pseudo-color image, for the rectangular block-like object, our method better restores the straight lines of the object's boundary, and there is no color fusion in areas where the color changes. In region 6, at the location pointed to by the orange arrow in the pseudo-color image, the lines reconstructed by other methods either show significant distortion and deformation or completely disappear in certain areas. However, the lines restored by our method are closer to those in the GT. From the error maps, darker regions indicate closer similarity to the GT. Therefore, in the areas marked by the red box in region 2 and the orange box in region 6, our proposed MIAN shows fewer bright white regions, which indicates that the image restored by our method is closer to the GT in many fine details.

For the Pavia Centre dataset, we present blocks from regions 2 and 4 of the test image in Fig. 5, using channels 11, 32, and 64 of the image to form pseudo-color images. The images from this dataset also demonstrate that our method achieves better performance. As indicated by the arrows on both error maps, the image reconstructed by our method has fewer bright spots, indicating that our image is closer to the GT.

To further demonstrate the effectiveness of the proposed method, we additionally conduct a single-band visualization experiment on several randomly selected test samples in the $\times 4$ setting. Specifically, we display the 20th spectral band of the third test image from the Chikusei dataset, the 51st band of the first test image from the Pavia Centre dataset, and the second band of the ninth test image from the CAVE dataset. For each case, we show the reconstructed single-band images and the corresponding residual maps for all compared methods. As illustrated in Fig. 6, MIAN produces noticeably

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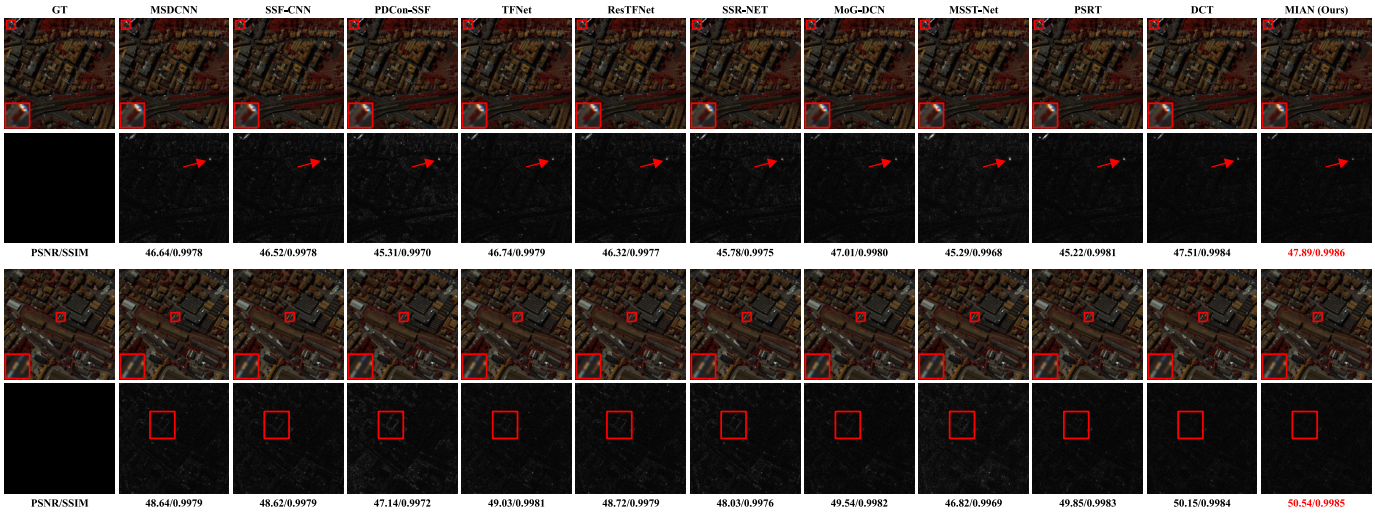


Fig. 5. Reconstructed pseudo-color images and error maps of various methods on selected regions of the Pavia Centre test set at $\times 4$ scale.

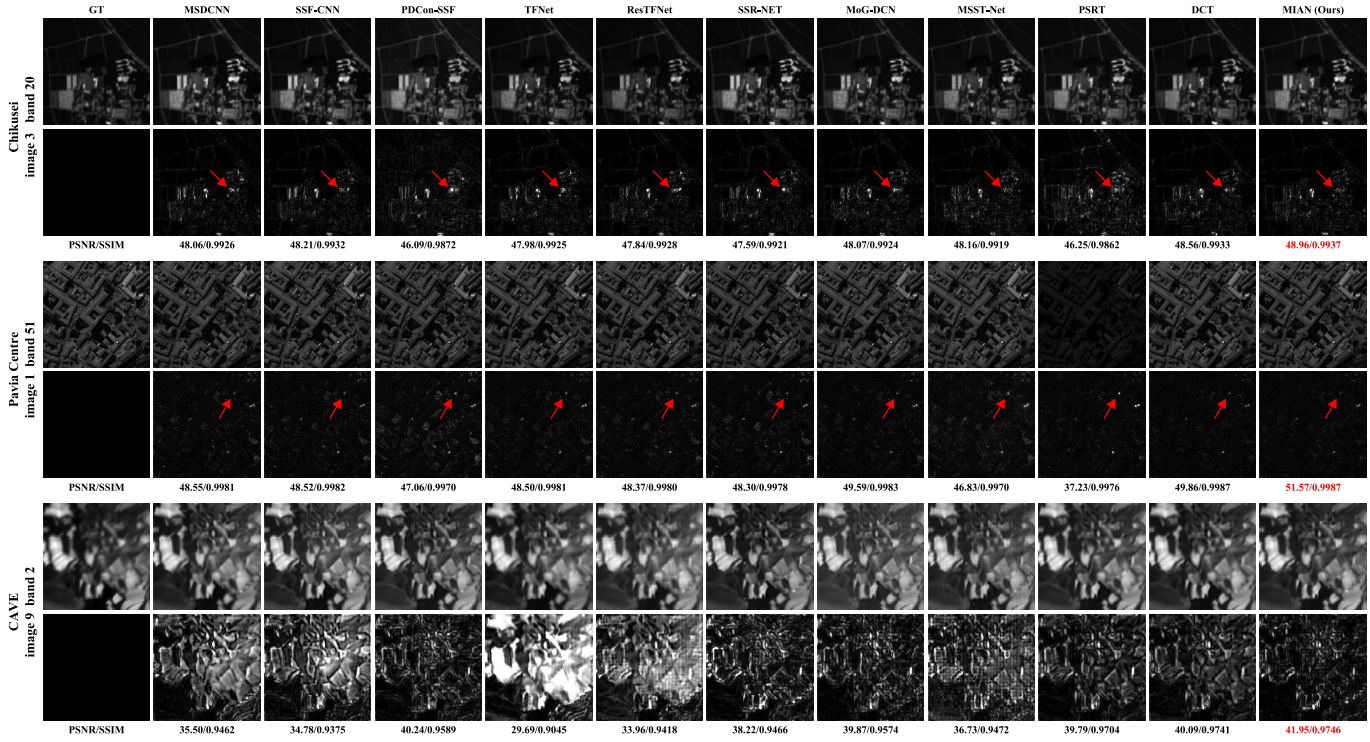


Fig. 6. Reconstructed single-band images and their corresponding error maps for selected bands from the Chikusei, Pavia Centre, and CAVE test sets at $\times 4$ scale.

smaller residuals and preserves fine structures more faithfully than both recent DL methods and traditional baselines. In the highlighted regions, the residual maps of MIAN contain fewer bright artifacts, indicating that its reconstructions are closer to the ground truth. These qualitative observations are consistent with the PSNR values reported for these bands.

c) Heatmaps: For the datasets used at $\times 32$ scale, we calculate the SAM error map and absolute error map between the GT and the images reconstructed by each method, and then plot the corresponding heatmaps in Fig. 7. The first two rows of the first column are the GT images for Chikusei and Pavia University test datasets. The first and third rows correspond to the SAM error maps, while the second and fourth rows correspond to the absolute error maps. For the Chikusei dataset, the absolute error heatmap is generated using the 60th spectral band, and for Pavia University, the 39th spectral band is used. From the color bars, it is evident that the lower the pixel value, the closer the image is to the blue color; conversely, higher values correspond to red. Since these are error maps, smaller values indicate that the method's reconstructed image is closer to the GT. From Fig. 7, we can observe that our method is consistently the bluest, indicating it is the closest to the GT. Additionally, in the areas of the image that are difficult to reconstruct (represented by green and red in the heatmap), our MIAN also performs the best in terms of reconstruction quality.

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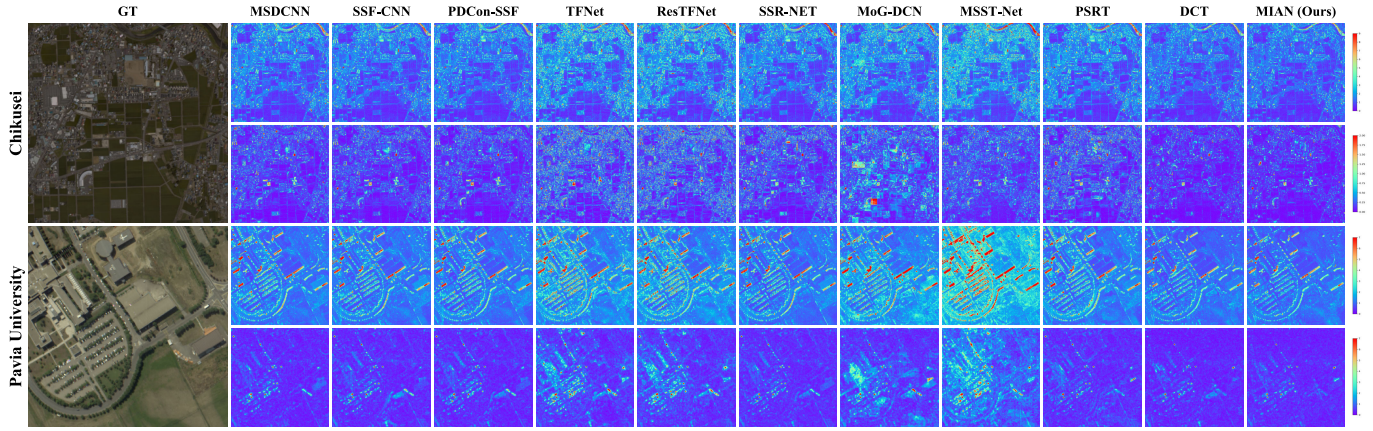


Fig. 7. SAM error map and absolute error map of different methods for Chikusei and Pavia University test images at ×32 scale.

TABLE IV
COMPARISON OF MODEL COMPLEXITY, COMPUTATIONAL COST, AND INFERENCE EFFICIENCY AMONG DIFFERENT METHODS

Method	MSDCNN	SSF-CNN	PDCon-SSF	TFNet	ResTFNet	SSR-NET	MoG-DCN	MSST-Net	PSRT	DCT	MIAN (Ours)
Params (M)	1.48	0.71	0.72	2.47	2.35	0.44	6.92	34.68	1.05	8.46	12.27
FLOPs (G)	24.21	11.61	6.88	8.65	8.01	7.26	212.43	32.33	18.19	332.15	519.68
Inference Times (ms)	1.97	0.40	0.5	2.24	2.53	0.75	38.32	24.12	15.24	321.54	345.5
PSNR (dB)	31.30	31.51	26.87	29.73	29.72	31.17	29.53	30.39	30.73	31.52	31.80

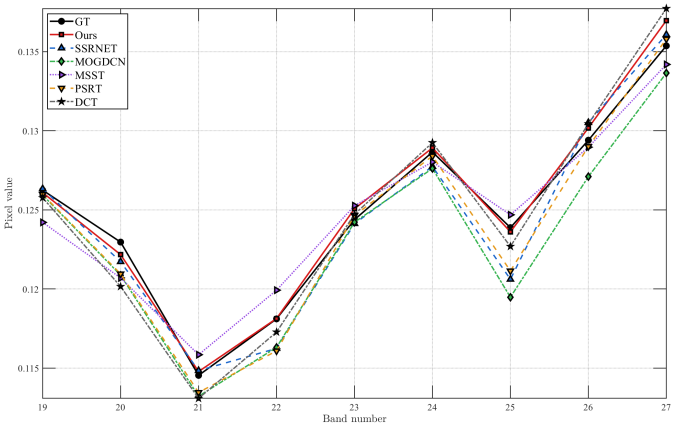


Fig. 8. Spectral response at location (94, 30) in the first test image of the CAVE dataset.

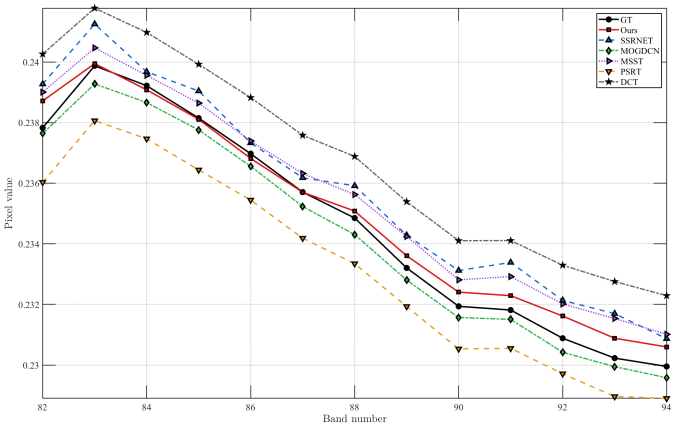


Fig. 9. Spectral response at location (35, 27) in the Chikusei test image.

d) Spectral response: We compare the ×4 scale fusion images generated by a selected set of representative recent methods (SSR-NET, MoG-DCN, MSST-Net, PSRT, DCT, and our MIAN). To improve readability, we visualize only the most informative spectral ranges for each dataset. For the Chikusei test image, we plot bands 82–94 at the pixel location (35, 27); for the Pavia Centre test image, bands 85–92 at location (144, 253) are visualized; and for the CAVE dataset, bands 19–27 at pixel location (94, 30) in the first test image are plotted. The corresponding spectral curves are shown in Figs. 8–10, respectively. As observed from the curves, MIAN produces spectral signatures that align most closely with the ground truth, demonstrating superior spectral reconstruction fidelity compared with the other methods.

e) Complexity analysis: We provide a comprehensive comparison of computational complexity across representative fusion models, including parameter count, floating point operations (FLOPs), and inference time, as summarized in Table IV. Although MIAN exhibits higher computational complexity than lightweight CNN-based baselines, its performance improvements are substantial rather than incremental. MIAN achieves markedly superior reconstruction accuracy, consistently outperforming both traditional methods and recent DL approaches across all datasets. This significant performance gain demonstrates that the additional computation introduced by our multiscale interaction architecture is meaningful and justified, providing clear evidence for the effectiveness of the proposed design.

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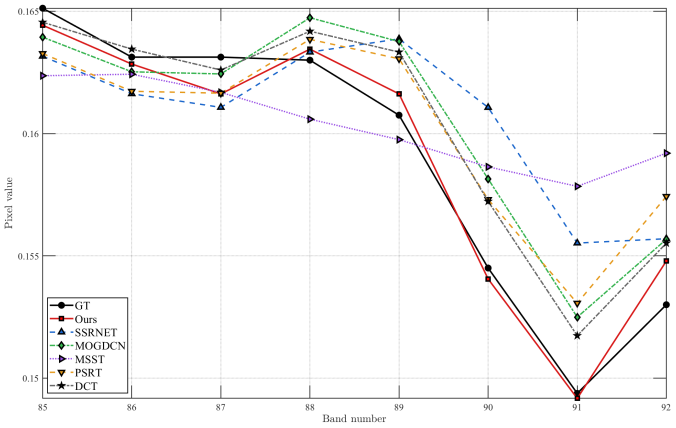


Fig. 10. Spectral response at location (144, 253) in the Pavia Centre test image.

TABLE V
COMPONENTWISE ABLATION EXPERIMENTS

Models	MIAB	WDAB	CAB	PSNR ↑	SAM ↓	ERGAS ↓
Baseline	✖	✖	✖	31.0707	4.3667	4.4039
Model-A1	✔	✖	✖	47.5550	1.8435	0.9676
Model-A2	✔	✔	✖	47.5563	1.8420	0.9672
Model-A3 (default)	✔	✔	✔	47.6573	1.8241	0.9587

f) Results on natural HSIs: Our model demonstrates excellent performance not only on remote sensing HSIs but also on natural images. To verify this, we evaluate the generalization ability of our model by randomly selecting three images from the CAVE dataset and comparing the reconstructed images from the competing methods and our MIAN. The selected images are "cd," "stuffed toy," and "jelly beans," using the 10th, 20th, and 31st spectral bands to construct pseudo-color images. These results are shown in Fig. 11, with the first, second, and third rows corresponding to the aforementioned images, respectively. We also calculate the PSNR values for each image compared to the GT. The first column displays the GT, the last column shows the images reconstructed by our method, and the intermediate columns represent the results from other methods. Our model achieves the best PSNR value for each image. From a visual perspective, in the first row, our model accurately reconstructed the green light in the top-left corner, whereas many other methods incorrectly generated red light.

2) Ablation Studies: In this section, we conduct a series of ablation experiments to explore the effectiveness of our model architecture and individual modules. The experiments are performed using the Pavia Centre dataset with a $\times 4$ spatial scale. For fair comparison, all models share the same implementation details.

a) Effect of each module: To assess the role of each architectural component, we perform ablation experiments by incrementally adding MIAB, WDAB, and CAB to the baseline model. As reported in Table V, the baseline model excludes all three modules. Introducing MIAB produces a significant improvement (Model-A1), as MIAB jointly exploits HR-MSI and LR-HSI inputs and models both spectral and spatial dependencies. Adding WDAB (Model-A2) further improves fusion quality by enlarging the effective receptive field and capturing cross-scale spatial structures. Incorporating CAB into all modules results in the full model (Model-A3), which achieves the best PSNR, SAM, and ERGAS values. The steadily increasing PSNR and decreasing SAM/ERGAS across the variants validate the contribution of each component.

TABLE VI
EFFECT OF NUMBER OF IDTS T

Models	T	PSNR ↑	SAM ↓	ERGAS ↓	SSIM ↑	RMSE ↓
Model-B1	1	47.6725	1.8226	0.9579	0.9974	0.0026
Model-B2	2	47.6799	1.8138	0.9574	0.9975	0.0026
Model-B3 (default)	3	47.8627	1.7904	0.9423	0.9975	0.0027

TABLE VII
EFFECT OF ATTENTION TYPES

Models	Attention Type	PSNR ↑	SAM ↓	ERGAS ↓
Model-C1	No Attention	35.1748	2.8963	2.8942
Model-C2	H2H Attention	36.0329	2.7003	2.6715
Model-C3	M2M Attention	46.9553	1.9075	1.0086
Model-C4	M2H Attention	47.4725	1.8528	0.9766
Model-C5	H2M Attention	47.2348	1.8683	0.9900
Model-C6	H2H Attention + H2H Attention	36.4553	2.6560	2.5598
MIAN (default)	H2M Attention + M2H Attention	47.7094	1.8170	0.9558

b) Effect of the number of IDTs T: In this section, we investigate the impact of the number of IDTs T on the performance of the model. Due to the complex structure inside the IDT, we limit its quantity to $\{1, 2, 3\}$, resulting in three models: Model-B1, Model-B2, and Model-B3. As shown in Table VI, the model's performance improves significantly when $T = 3$. Therefore, we set the number of IDTs to 3 in our final configuration.

c) Effect of attention types: This part investigates the impact of different attention types on model performance. Model-C1 is a network variant that does not use any attention mechanism. Model-C2 and Model-C3 correspond to self-attention using only HSI features and MSI image features, respectively. Model-C4 and Model-C5 represent interaction attention from MS to HS and from HS to MS, respectively. Model-C6 employs self-attention on HS features using two branches. Our proposed MIAN utilizes both the MS-to-HS interaction attention and HS-to-MS interaction attention. The performance metrics of these model variants are presented in Table VII.

From Table VII, it can be observed that the model's performance improves significantly when interaction attention is used. When both MS-to-HS and HS-to-MS interaction attention mechanisms are employed, the model achieves optimal performance. Therefore, we ultimately adopt this interaction attention as part of our model.

d) Effect of number of MIABs M: MIAB is a core module in our network, and its quantity significantly influences the

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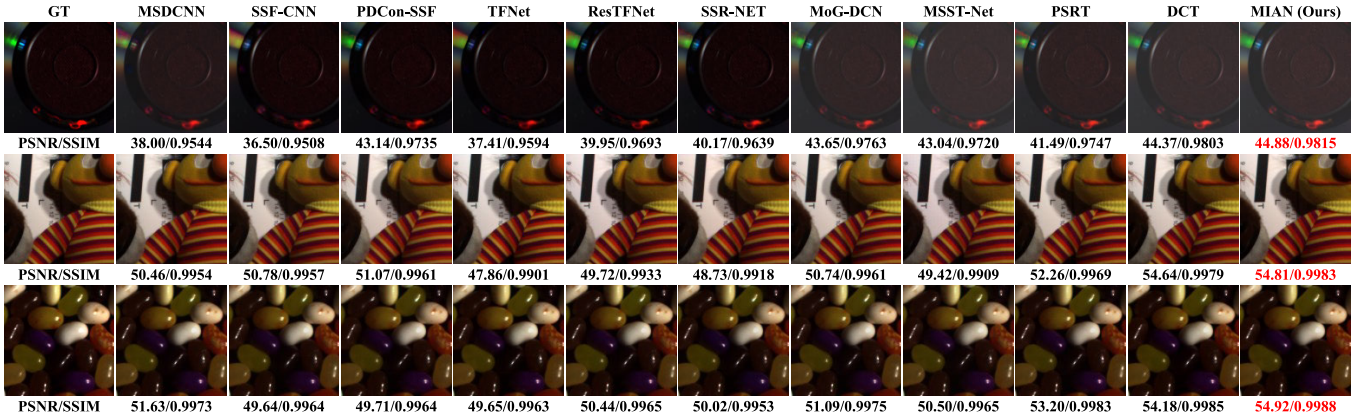


Fig. 11. Pseudo-color images and PSNR values of three selected images from the CAVE test set reconstructed by different methods.

TABLE VIII
EFFECT OF NUMBER OF MIABS M

Models	M	PSNR ↑	SAM ↓	ERGAS ↓	SSIM ↑	RMSE ↓
Model-D1	2	47.3995	1.8601	0.9790	0.9973	0.0027
Model-D2	4	47.5110	1.8451	0.9714	0.9974	0.0027
Model-D3 (default)	6	47.7081	1.8172	0.9555	0.9975	0.0026
Model-D4	8	47.8391	1.7956	0.9447	0.9975	0.0026

TABLE IX
EFFECT OF WINDOW SIZE

Models	Window Size	PSNR ↑	SAM ↓	ERGAS ↓	SSIM ↑	RMSE ↓
Model-E1	4×4	47.7029	1.8179	0.9555	0.9975	0.0026
Model-E2 (default)	8×8	47.7211	1.8167	0.9541	0.9975	0.0026
Model-E3	16×16	47.7397	1.8155	0.9537	0.9975	0.0026

TABLE X
EFFECT OF DILATION FACTOR γ

Models	γ	PSNR ↑	SAM ↓	ERGAS ↓	SSIM ↑	RMSE ↓
Model-F1	0	47.6773	1.8226	0.9580	0.9974	0.0026
Model-F2	0.3	47.6799	1.8200	0.9579	0.9975	0.0026
Model-F3 (default)	0.5	47.7111	1.8197	0.9567	0.9975	0.0026
Model-F4	0.8	47.6898	1.8186	0.9579	0.9975	0.0026

model's performance. To investigate this, we conduct multiple experiments by setting the number of MIABs M to different values from $\{2, 4, 6, 8\}$ and observe its impact on the model's performance. Table VIII presents the trends in PSNR, SAM, and ERGAS for the MIAN model on the Pavia Centre dataset as M changes.

As shown in Table VIII, the model's performance improves progressively with an increasing number of MIABs. However, considering the trade-off between model complexity and performance, we set the default number of MIABs to 6.

e) Effect of window size: Due to the characteristics of the attention mechanism, larger window sizes play a crucial role in capturing internal feature information over a larger range. However, the computational complexity of attention grows quadratically as the number of input tokens increases. Thus, how to choose an optimal window size becomes an important issue. We select the window sizes from $4, 8, 16$ and construct three model variants accordingly.

According to the data in Table IX, as the attention window size increases, the model's performance gradually improves. However, when the window size reaches 16, the performance improvement becomes more gradual, showing signs of saturation. From the trade-off between model complexity and performance, the window size of 8 provides near-optimal performance with relatively light computational burden. Therefore, we choose to set the window size to 8.

f) Effect of dilation factor γ: The dilation operation in WDAB enables the image patches for attention computation to expand further. We set the dilation factor γ to different values from $\{0, 0.3, 0.5, 0.8\}$ to observe its impact on model performance. Here, $\gamma = 0$ means no dilation operation, equivalent to using standard window attention for computation. Meanwhile, $\gamma = 0.3$, $\gamma = 0.5$, and $\gamma = 0.8$ correspond to padding one, two, and three layers of pixels around the image patches, respectively, when the window size is set to 8. Because the padding width is computed as and must be an integer number of pixels, only a discrete set of γ values leads to valid dilated windows under a fixed window size. These discrete γ values expand the receptive region progressively, enabling different levels of contextual aggregation.

As shown in Table X, as γ increases, the model's PSNR value initially rises and then falls, while the ERGAS value first decreases and then increases. With the dilation operation extending the query window, the model gains a larger receptive field and captures more contextual information, leading to an improvement in PSNR. However, an excessively large query window prevents attention from focusing on key information and causes significant mismatches with the key and value windows, ultimately reducing the PSNR. Therefore, the best performance is achieved when $\gamma = 0.5$. Consequently, we set the dilation factor to 0.5.

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V. Conclusion

In this article, we propose a MIAN for HS-MS- remote sensing image fusion. Our approach focuses on extracting key information from both modalities. Within the network architecture, we introduce an MIAB and a WDAB. The former effectively extracts, coordinates, and fuses the information from both modalities, aligning the spatial and channel correlations with the distribution of real-world remote sensing images. The latter overcomes the limited field of view of traditional window attention by expanding the receptive field and capturing more cross-scale self-similarity information for fusion. Additionally, we integrate CABs into the attention modules to address the high dimensionality of HS channels, which is challenging to process. As a result, our proposed network achieves significant improvements in both objective metrics and subjective visual quality across multiple HS datasets. Furthermore, ablation studies demonstrate the effectiveness of each module in our network.

There remain many challenges to address in future work. For example, integrating our method with image alignment, unsupervised learning strategies, or other techniques to tackle broader problems. Additionally, extending our method to applications such as single-image HS super-resolution and pansharpening could be promising directions.

References

[1] Y. Yang, J. Qu, W. Dong, T. Zhang, S. Xiao, and Y. Li, "TMCFN: Text-supervised multidimensional contrastive fusion network for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511015.

[2] J. Qu, W. Dong, Y. Yang, T. Zhang, Y. Li, and Q. Du, "Cycle-refined multidecision joint alignment network for unsupervised domain adaptive hyperspectral change detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 2, pp. 2634–2647, Feb. 2025.

[3] S. Jia, L. Shen, J. Zhu, and Q. Li, "A 3-D Gabor phase-based coding and matching framework for hyperspectral imagery classification," *IEEE Trans. Cybern.*, vol. 48, no. 4, pp. 1176–1188, Apr. 2018.

[4] J. Qu, J. Zhao, W. Dong, S. Xiao, Y. Li, and Q. Du, "Feature mutual representation-based graph domain adaptive network for unsupervised hyperspectral change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5501214.

[5] R. Dian, T. Shan, W. He, and H. Liu, "Spectral super-resolution via model-guided cross-fusion network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 10059–10070, Jul. 2024.

[6] L. Tan, R. Dian, S. Li, and J. Liu, "Frequency-spatial domain feature fusion for spectral super-resolution," *IEEE Trans. Comput. Imag.*, vol. 10, pp. 589–599, 2024.

[7] H. Ghassemian, "A review of remote sensing image fusion methods," *Inf. Fusion*, vol. 32, pp. 75–89, Nov. 2016.

[8] S. Fang and H. Gan, "SSUN-Net: Spatial–spectral prior-aware unfolding network for pan-sharpening," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 3, pp. 2897–2905.

[9] M. Selva, B. Aiazzi, F. Butera, L. Chiarantini, and S. Baronti, "Hyper-sharpening of hyperspectral data: A first approach," in *Proc. 6th Workshop Hyperspectral Image Signal Processing: Evol. Remote Sens. (WHISPERS)*, Jun. 2014, pp. 1–4.

[10] X. Wang, Q. Hu, Y. Cheng, and J. Ma, "Hyperspectral image super-resolution meets deep learning: A survey and perspective," *IEEE/CAA J. Autom. Sinica*, vol. 10, no. 8, pp. 1668–1691, Aug. 2023.

[11] N. Yokoya, T. Yairi, and A. Iwasaki, "Coupled nonnegative matrix factorization unmixing for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 528–537, Feb. 2012.

[12] M. Simões, J. Bioucas-Dias, L. B. Almeida, and J. Chanussot, "A convex formulation for hyperspectral image superresolution via subspace-based regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3373–3388, Jun. 2015.

[13] C. Lanaras, E. Baltsavias, and K. Schindler, "Hyperspectral super-resolution by coupled spectral unmixing," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 3586–3594.

[14] R. Dian, L. Fang, and S. Li, "Hyperspectral image super-resolution via non-local sparse tensor factorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3862–3871.

[15] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[16] R. Dian, S. Li, A. Guo, and L. Fang, "Deep hyperspectral image sharpening," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 11, pp. 5345–5355, Nov. 2018.

[17] Q. Xie, M. Zhou, Q. Zhao, D. Meng, W. Zuo, and Z. Xu, "Multispectral and hyperspectral image fusion by MS/HS fusion net," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1585–1594.

[18] W. Wang, W. Zeng, Y. Huang, X. Ding, and J. Paisley, "Deep blind hyperspectral image fusion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4149–4158.

[19] W. Dong, C. Zhou, F. Wu, J. Wu, G. Shi, and X. Li, "Model-guided deep hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 5754–5768, 2021.

[20] J.-F. Hu, T.-Z. Huang, L.-J. Deng, H.-X. Dou, D. Hong, and G. Vivone, "Fusformer: A transformer-based fusion network for hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[21] S.-Q. Deng, L.-J. Deng, X. Wu, R. Ran, D. Hong, and G. Vivone, "PSRT: Pyramid shuffle-and-reshuffle transformer for multispectral and hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503715.

[22] X. Cao, Y. Lian, K. Wang, C. Ma, and X. Xu, "Unsupervised hybrid network of transformer and CNN for blind hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5507615.

[23] Q. Wei, J. Bioucas-Dias, N. Dobigeon, and J.-Y. Tourneret, "Hyperspectral and multispectral image fusion based on a sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3658–3668, Jul. 2015.

[24] Q. Wei, N. Dobigeon, and J.-Y. Tourneret, "Bayesian fusion of hyperspectral and multispectral images," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2014, pp. 3176–3180.

[25] R. A. Borsoi, T. Imbiriba, and J. C. M. Bermudez, "Super-resolution for hyperspectral and multispectral image fusion accounting for seasonal spectral variability," *IEEE Trans. Image Process.*, vol. 29, pp. 116–127, 2020.

[26] W. Dong et al., "Hyperspectral image super-resolution via non-negative structured sparse representation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2337–2352, May 2016.

[27] C.-H. Lin, F. Ma, C.-Y. Chi, and C.-H. Hsieh, "A convex optimization-based coupled nonnegative matrix factorization algorithm for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1652–1667, Mar. 2018.

[28] Y. Chen, J. Zeng, W. He, X.-L. Zhao, and T.-Z. Huang, "Hyperspectral and multispectral image fusion using factor smoothed tensor ring decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515417.

[29] C. Prévost, P. Chainais, and R. Boyer, "Fast fusion of hyperspectral and multispectral images: A tucker approximation approach," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2022, pp. 2076–2080.

[30] S. E. Brezini and Y. Deville, "Hyperspectral and multispectral image fusion with automated extraction of image-based endmember bundles and sparsity-based unmixing to deal with spectral variability," *Sensors*, vol. 23, no. 4, p. 2341, Feb. 2023.

[31] R. A. Borsoi, C. Prévost, K. Usevich, D. Brie, J. C. M. Bermudez, and C. Richard, "Coupled tensor decomposition for hyperspectral and multispectral image fusion with inter-image variability," *IEEE J. Sel. Topics Signal Process.*, vol. 15, no. 3, pp. 702–717, Apr. 2021.

[32] Y. Chang, L. Yan, H. Fang, S. Zhong, and W. Liao, "HSI-DeNet: Hyperspectral image restoration via convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 667–682, Feb. 2019.

[33] Q. Yuan, Y. Wei, X. Meng, H. Shen, and L. Zhang, "A multiscale and multidepth convolutional neural network for remote sensing imagery pan-sharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 978–989, Mar. 2018.

[34] X. Liu, Q. Liu, and Y. Wang, "Remote sensing image fusion based on two-stream fusion network," *Inf. Fusion*, vol. 55, pp. 1–15, Mar. 2020.

[35] X.-H. Han, B. Shi, and Y. Zheng, "SSF-CNN: Spatial and spectral fusion with CNN for hyperspectral image super-resolution," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 2506–2510.

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[36] X. Zhang, W. Huang, Q. Wang, and X. Li, "SSR-NET: Spatial–spectral reconstruction network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5953–5965, Jul. 2021.
[37] R. Dian, S. Li, and L. Fang, "Learning a low tensor-train rank representation for hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2672–2683, Sep. 2019.
[38] M. Zhou, J. Huang, Y. Fang, X. Fu, and A. Liu, "Pan-sharpening with customized transformer and invertible neural network," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, 2022, pp. 3553–3561.
[39] J.-F. Hu, T.-Z. Huang, L.-J. Deng, T.-X. Jiang, G. Vivone, and J. Chanussot, "Hyperspectral image super-resolution via deep spatiospectral attention convolutional neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 7251–7265, Dec. 2022.
[40] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Aug. 2020.
[41] S. Jia, Z. Min, and X. Fu, "Multiscale spatial–spectral transformer network for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 96, pp. 117–129, Aug. 2023.
[42] Q. Ma, J. Jiang, X. Liu, and J. Ma, "Reciprocal transformer for hyperspectral and multispectral image fusion," *Inf. Fusion*, vol. 104, Apr. 2024, Art. no. 102148.
[43] W. Dong, S. Liu, S. Xiao, J. Qu, and Y. Li, "ISPDiff: Interpretable scale-propelled diffusion model for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519614.
[44] J. Fang, J. Yang, A. Khader, and L. Xiao, "MIMO-SST: Multi-input multi-output spatial–spectral transformer for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5510020.
[45] J. Li, S. Du, R. Song, Y. Li, and Q. Du, "Progressive spatial information-guided deep aggregation convolutional network for hyperspectral spectral super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 1, pp. 1677–1691, Jan. 2025.
[46] K. Zhang et al., "Spectral–spatial dual graph unfolding network for multispectral and hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508718.
[47] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.
[48] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[49] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.
[50] N. Yokoya and A. Iwasaki, "Airborne hyperspectral data over Chikusei," Dept. Space Application Laboratory, Univ. of Tokyo, Tokyo, Japan, Tech. Rep. SAL-2016-05-27, May 2016.
[51] F. Dell'Acqua, P. Gamba, A. Ferrari, J. A. Palmason, J. A. Benediktsson, and K. Arnason, "Exploiting spectral and spatial information in hyperspectral urban data with high resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 1, no. 4, pp. 322–326, Oct. 2004.
[52] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Postcapture control of resolution, dynamic range, and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.
[53] C. Yi et al., "Aerial hyperspectral remote sensing classification dataset of Xiongan new area (matiwan village)," *Nat. Remote Sens. Bull.*, vol. 24, no. 11, pp. 1299–1306, 2020.
[54] L. Wald, T. Ranchin, and M. Mangolini, "Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images," *Photogrammetric Eng. Remote Sens.*, pp. 691–699, 1997.
[55] X. Cao, Y. Lian, Z. Liu, J. Wu, W. Zhang, and J. Liu, "Hyperspectral image super-resolution via spectral matching and correction," *J. Opt. Soc. Amer. A, Opt. Image Sci.*, vol. 40, no. 8, pp. 1635–1643, 2023.
[56] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
[57] F. A. Kruse et al., "The spectral image processing system (SIPS)—Interactive visualization and analysis of imaging spectrometer data," *Remote Sens. Environ.*, vol. 44, nos. 2–3, pp. 145–163, May 1993.
[58] L. Wald, "Quality of high resolution synthesised images: Is there a simple criterion," in *Proc. 3rd Conf. Fusion Earth Data, Merging Point Meas.*, 2014, pp. 99–103.
[59] X. Zhang, "Validation and correlation analysis of metrics for evaluating performance of image fusion," *Acta Autom. Sinica*, pp. 306–315, 2014.



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